```
address: akka.actor.Address = akka://words@127.0.0.1:2551
scala> Cluster(guardian).manager ! Down(address)
Cluster Node [akka://words@127.0.0.1:2552] - Marking unreachable node
➥[akka://words@127.0.0.1:2551] as [Down]                                    ❶
Cluster Node [akka://words@127.0.0.1:2552] - Leader can perform its duties
➥again                                                                      ❷
Cluster Node [akka://words@127.0.0.1:2552] - Leader is removing unreachable
➥node [akka://words@127.0.0.1:2551]                                         ❸
Association to [akka://words@127.0.0.1:2551] with UID [3375074971115939575] is
irrecoverably failed. UID is now quarantined and all messages to this UID will be
delivered to dead letters. Remote ActorSystem must be restarted to recover from this
situation. Reason: Cluster member removed, previous status [Down]
❹
```

❶ Seed node 1 is set to Down.

❷ Seed node 2 can now set any member to Up, Down, Exiting, or Removed.

❸ Seed node 1 is set to Removed.

❹ Seed node 1 is quarantined and removed.

The output also shows that the `ActorSystem` of seed node 1 would have to reboot if it wanted to reconnect. Although an unreachable node can be configured via settings to shut down automatically after a certain time, this is not recommended. It is best to use the Akka SBR, which will automatically make this decision for you based on your configuration. You can set it by adding the following to your .conf:

```
akka.cluster.downing-provider-class =
➥"akka.cluster.sbr.SplitBrainResolverProvider"
```

This topic is discussed in more detail in chapter 13.

Failure detector

The cluster module uses an implementation of a *ϕ accrual failure detector* to detect unreachable nodes. The work is based on a paper by Naohiro Hayashibara, Xavier Défago, Rami Yared, and Takuya Katayama